PowerCo Energy Demand, London

# November 2025

# Minimising Exposure to London demand volatility

PREPARED BY THE
Data Management Department
Team 11 (London)

# Data Management Team 11, London

# TASK FORCE ON ENERGY DEMAND PREDICTION MODELING

---

**Members and Contributors:**
Christie Chong
Edman Ho
Effie Armah-Tetteh
Gaspar Da Silveira
Giovanna Lima
Neelabjo Roy

## CONTENTS

1. INTRODUCTION....1
    1.1 Background....1
    1.2 Problem Definition....1
2. METHODOLOGY....1
    2.1 Data Acquisition....1
    2.2 Data Cleaning....2
    2.3 Exploratory Data Analysis....2
    2.4 Data Modelling....3
    2.5 Model Interpretation and Results....3
    2.6 Model Improvement....4
3. REFLECTIONS....4
4. REFERENCES....5
    A.1 PostgresSQL Database....5
    A.2 Pearson Correlation Matrix....5
    A.3 Seasonal Demand 2022 – 2025....6
    A.4Actual vs. SARIMA demand....6

# 1. INTRODUCTION

## 1.1 Background

Energy demand forecasting is foundational to maintain the stability, reliability, and economic efficiency of modern power systems. As energy grids become more dynamic and sensitive to short-term fluctuations, understanding how external factors affect energy consumption has become increasingly important. The interplay between meteorological factors (temperature, solar radiation, precipitation, wind) and human behaviour results in complex patterns that traditional forecasting approaches often struggle to capture.

This project, developed by PowerCo Data Management Team 11 (London), aims to build a comprehensive end-to-end pipeline capable of cleaning, integrating, analysing, and modelling energy demand using both historical consumption and weather variables. PowerCo can rely on accurate forecasting to schedule generation, plan supply, manage reserves, and prevent imbalances.

## 1.2 Problem Definition

PowerCo (London) seeks to understand:

- How do weather conditions influence daily energy demand?
- Which meteorological factors contribute most to demand variability?
- Can a robust predictive model improve forecasting accuracy?
- How should diverse datasets be integrated and managed for effective analysis?

Thus, the business challenge is a data integration and prediction problem: **designing a scalable, reliable data architecture and modelling approach capable of forecasting daily demand using weather and temporal signals.**

# 2. METHODOLOGY

## 2.1 Data Acquisition

The data was drawn from two main sources: the National Energy System Operator Portal for data on historic energy demand and weather data from Open Meteo. To create a unified data frame for historic energy demand, we combined tables from 2015 to 2022 together.

We acquired weather data by calling two tables from the Open Meteo API: the historical weather forecast table and the historical weather table. Since we were predicting daily demand, we selected additional weather features that were

aggregated by day, including shortwave radiation, precipitation, and windspeed. To create a data frame, we extracted the data called from the API using the "daily" key and transformed the data into separate data frames.

## 2.2 Data Cleaning

The date format was different for both tables. Thus, we used a function to transform all the dates in the energy demand table so that all the dates were recorded using the ISO date format. Once the data was cleaned and pre-processed, we stored the weather datasets and the energy demand datasets into a Postgres SQL server for full team accessibility.[1]

Additionally, to consider holiday factors, we created a table in the server that recorded different calendar events such as bank holidays. Once all the tables were created, we joined the tables together using the settlement date column to create a unified data frame ready for modelling.

We then queried the database to create two tables; we first created a daily demand table, featuring: date, daily demand, average daily demand, and daily peak demand. We then joined the daily demand to the weather forecast table, and separately to the observed weather table. Each of the resulting joined tables had the same shape because they consider the same time periods; this was useful in that we could compare the effectiveness of predicting demand with observed weather data vs. forecasted weather data with all other variables controlled.

## 2.3 Exploratory Data Analysis[2]

Before modelling the data, we conducted EDA to further understand each feature. We plotted histograms for several key features and noticed that short-wave radiation, gust wind speed, and precipitation hours were right-skewed while other features were mostly symmetrical and roughly normally distributed.

We also plotted correlation heatmaps to understand which variables were the most correlated with daily average energy demand. Through the heatmap, we noticed that temperature, apparent temperature, and shortwave radiation have strong negative correlations with energy demand. From these results, we can conclude that temperature, radiation, and peak demand are strong drivers of daily demand, while wind and precipitation show much weaker effects. This is further supported by the scatter plots we created that mapped individual variables with daily average energy demand. Temperature and demand had especially strong inverse relations with demand.

Moreover, we plotted energy demand over multiple years to see whether there were any seasonal factors that would impact our modelling process. We observed that energy demand peaked in Winter and experienced a trough in Summer: we made a logical guess that people were more likely to turn on heaters and use more energy during Winter. We also noticed that energy demand was more subdued during weekends.

[1] See appendix A.1
[2] See appendices A.2 and A.3

## 2.4 Data Modelling

We experimented with multiple models to see which models best predicted the data. Since the data was a time series, it was not appropriate to assign each observation randomly to a test or train set. Instead, we split the data into train and test sets using settlement date. Training data used observations where the settlement period was before 2023 while testing set used data starting from January 1st 2023.

### Linear model

We first created an OLS model as a baseline to compare our models with. This linear model predicted energy demand using only average temperature. This model did not perform well since its $R^2$ was only 0.6 on the test set.

Considering the seasonality observed during EDA, we created an additional OLS model that predicted energy demand against using time lagged variables and additional weather variables. Although the $R^2$ was 0.86, the multi linear regression model experienced first order autocorrelation issues since the Durbin Watson test statistic came out to be 1.2, which is very far from the ideal value of 2. Hence, we needed to use a different model to fit the data.

### SARIMA model

We chose to use a SARIMA model to predict energy demand since SARIMA could capture seasonality and maintain interpretability. To determine the parameters, we plotted an ACF and a PACF plot, observing that there was a seasonal cycle that lasted until lag=15. The ACF plot also exhibited a gradual decay; thus, differencing may be needed too. Hence, we chose to set our seasonal order to 15 and p=1. We experimented with differencing greater than 0, but the model failed to represent the test observations well. Tuning parameter q, we found that using q=2 yielded the best results. We also used weather data and lagged weather data to capture weather impacts on energy usage. As a result, the model achieved an $R^2$ value of 0.94 on the training set.

Checking our model with the Durbin Watson test, we noticed that the test statistic came out to be 1.92 for the SARIMA model, which meant that there were no significant first-order autocorrelation issues with the model. We also tested other model assumptions such as whether residuals were normally distributed by plotting a QQ-plot and whether the residuals were independent by plotting residuals against time.

## 2.5 Model Interpretation and Results[3]

Our model prediction was mostly dominated by additional weather and calendar date regressors. Temperature, shortwave radiation, and lagged temperature played a significant role in predicting energy usage. These variables can be interpreted as a person's own personal experience with how warm it is on a given day or during the day before. As such, how the cold or warm a person felt the day before can predict whether they turn on the heater the next day.

[3] See appendix A.4

Day of the week and weekends also played a role in predicting energy usage: people were more likely to stay inside during weekends. However, whether bank holidays were not a significant predictor of energy usage in our model. Based on the ACF plot, we can also see that energy demand goes through a bi-weekly cycle, potentially pay day or benefits cycles.

### 2.6 Model Improvement

To further improve our model, we could incorporate text embeddings in the features we used to predict energy usage. One way to do so is to incorporate the sentiment of weather news reports into our model. People may be more inclined to turn on the heating or turn on the AC when extreme language is used to describe the weather. Hence, it may help predict how much energy will be used.

We could also use a different model to better capture the decay observed in the ACF plot. Although we attempted to add differencing to the SARIMA model by setting d>0, it reduced model fit on the test data with an $R^2$ of 0.38. The ARIMA component of our model was also making many time series corrections, further indicating that our current model is still influenced by time series correlations in the data. Since our model was created using SARIMA, it is not as good as other models in predicting longer term energy demand and easily reverts to predicting the mean when forecasting on a longer time horizon.

Additionally, we could also improve our model by finding a better way to capture seasonality. We could have encoded using a seasonal dummy variable by mapping specific months with seasons rather than relying on bi-weekly cycles and used it as an exogenous variable. Alternatively, we could use a different model since the SARIMA model cannot handle multiple periodicities and multiple seasonalities. By incorporating more complex seasonalities, we can further improve our predictions of energy demand.

## 3. REFLECTIONS

The SARIMA model has successfully predicted the energy demand on both testing and training data, with 94% accuracy. This is a significant improvement to PowerCo's current demand-supply match rate; this is only 73%.[4] By thoroughly exploring our database and the relationships between demand and external factors, we confirmed that Summer and Winter demand is stable each year while Autumn and Spring are prone to more short-term fluctuations. Nonetheless, the model can predict fluctuations by analysing weather data from the previous 15-day demand cycle.

By using the model to reduce under/oversupply of energy and the added storage costs (£0.47m/annum), PowerCo is predicted to reduce its annual operating costs by £3.12m per annum.[5]

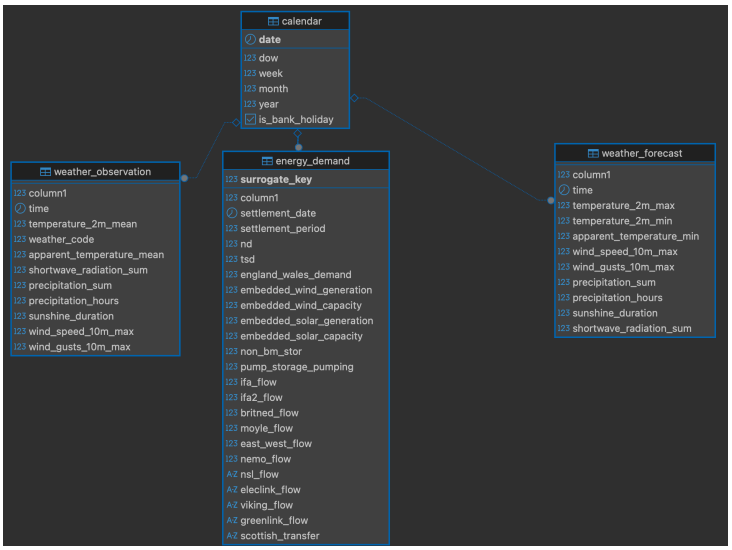
[4] This figure is completely made up!
[5] Ibid.

# 4. REFERENCES

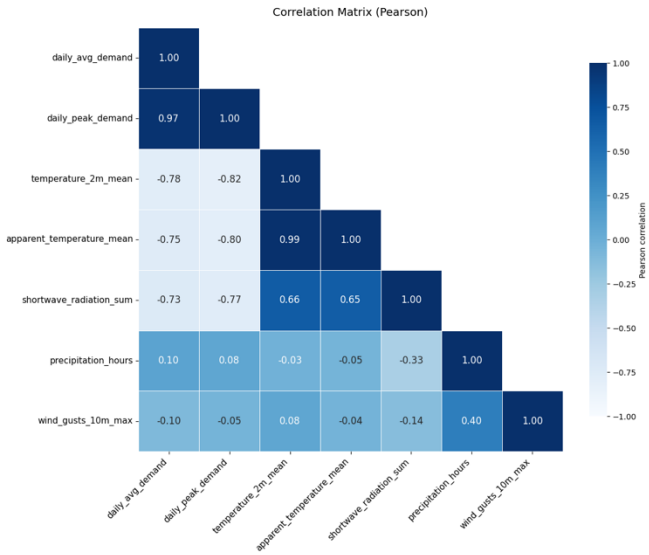

1. **New England States Energy Office.** Data Portal. Available at: https://www.neso.energy/data-portal (accessed November 17, 2025).
2. **Open-Meteo.** Weather API – Free Weather Data for Research and Developers. Available at: https://open-meteo.com/ (accessed November 17, 2025).


## APPENDIX A

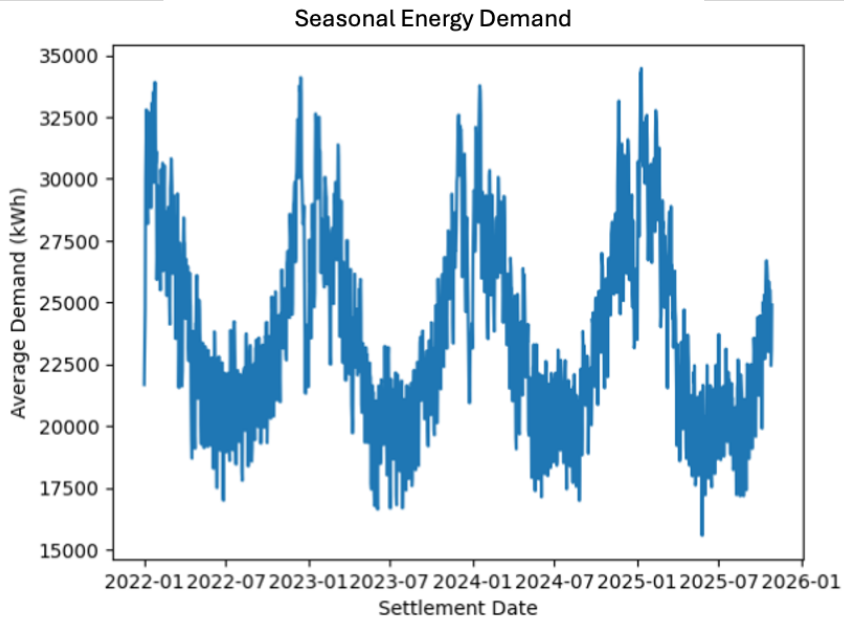
### A.1 PostgresSQL Database

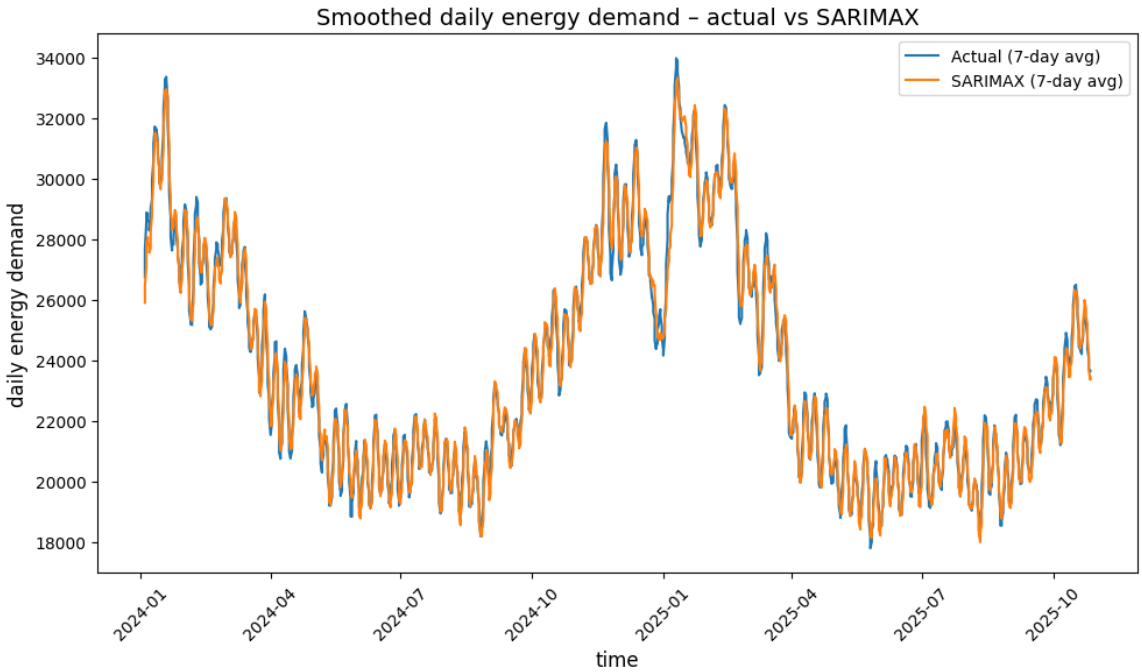



### A.2 Pearson Correlation Matrix



## A.3 Seasonal Demand 2022 – 2025




## A.4 Actual vs. SARIMA demand